

Perceived Income Risk Across the Income Distribution

1 Summary

In the New York Fed’s Survey of Consumer Expectations (SCE), workers in households earning under \$50k report wage uncertainty roughly two and a half times that of workers in households above \$100k. Their uncertainty beliefs are also stickier, and they revise them more after realised wage shocks. Matched same-job spells in the Survey of Income and Program Participation (SIPP) show no comparable gradient in the realised wage process. The gap sits in beliefs, not in wages.

The argument in three steps.

1. *The belief gradient is not a wage gradient* (Sections 3–4). Conditional on remaining at the same job, perceived wage risk falls steeply with income, is most persistent at the bottom, and co-moves with realised shocks. The realised process is flat to mildly *increasing* in income. The standard heterogeneous-agent calibration, which sets perceived risk equal to realised risk (the FIRE benchmark), therefore fails on all three margins at once.
2. *One learning rule organises all three deviations* (Sections 5–7). Agents learn the variance of their permanent earning power from realised pay signals. The rule has three interpretable parameters: how fast beliefs update, what share of each shock is read as permanent, and a prior on transitory risk. Estimated within person by income group, the bottom updates slowest, reads three times as much of each shock as permanent, and holds a prior almost four times its realised transitory risk. The middle and top sit near each other and near their realised processes. The persistence and prior gradients are statistically significant; the attribution gradient is monotone but not.
3. *My proposed microfoundation: the wedges come from how pay is delivered, not how risky it is* (Sections 8–10). Pay at the bottom arrives in noisy fragments and largely unlabelled; bonuses and commissions at the top arrive as marked line items. A Bayesian agent facing this signal structure reproduces all three estimated patterns (prior and persistence in levels, attribution in ratios), without requiring any sort of behavioural bias. Embedded in an Aiyagari model, I show that the calibrated beliefs open precautionary-saving and marginal-propensity-to-consume (MPC) gradients that a FIRE calibration cannot generate; the wedge falls on the middle and top, while the bottom, holding the most distorted beliefs but near hand-to-mouth, is moved by liquidity rather than by its own beliefs.

Relation to the literature. Wang (2023) documents the level gap between perceived and calibrated income risk in the SCE; I add its cross-sectional structure, its within-person dynamics, an estimated updating rule, and a mechanism. Rozsypal and Schlafmann (2023) document income-

graded bias in expected income *levels*; I study the second moment. Caplin et al. (2023) find perceived earnings risk below econometric estimates in Danish register-linked data, consistent with my level finding. Kaufmann and Pistaferri (2009) show superior information rationalises part of such gaps; my labelled-pay channel is that information channel made observable. Broer et al. (2022) study information heterogeneity in heterogeneous-agent settings. The mechanism builds on documented compensation facts: performance pay concentrates at the top (Lemieux et al., 2009), high-income shocks are predominantly transitory (Guvenen et al., 2021), bottom-of-distribution income is dominated by volatile, fragmented hours and gig flows (Hardy and Ziliak, 2014; Morduch and Schneider, 2017), and recessions hit the bottom hardest (Hoynes et al., 2012).

Status. The empirics (Sections 3–7, Appendices A–F) are complete. The mechanism of Section 8 is so far only calibrated, but it is directly testable. A partial-equilibrium implementation of the Section 9 model runs; general equilibrium is future work.

2 Data and measurement

Samples. The SIPP sample covers reference years 2013–2023 and the SCE runs 2013–2025; both are monthly within-respondent panels. I restrict to same-job, same-employer spells of at least 6 consecutive months and apply survey weights throughout. Respondents are sorted into three income groups by median annual household income: Bot (<\$50k), Mid (\$50k–\$100k), Top ($\geq$ \$100k).¹

SCE. The density question asks for the distribution of next-12-month earnings growth “*on the same job, same hours, same employer*”, on a 10-bin histogram. The object I take from it, perceived variance $\bar{\sigma}_{\text{perc}}^2$, is the bin-midpoint variance of each respondent’s reported histogram. Respondents placing all mass in a single bin are excluded, and the result is trimmed at 1%/99%. I residualise on age, sex, state, education, survey tenure, and year-month fixed effects; survey tenure absorbs panel conditioning (Kim and Binder, 2023). I drop each respondent’s observations from the first wave in which they report a job change.

SIPP. Per-month earnings include regular wage and salary plus bonuses, commissions, tips, and overtime; the self-employed are excluded. I residualise log wages on demographics, state, and year-month fixed effects and trim squared changes at 1%/99%.

3 Perceived risk is higher and more persistent at the bottom

Levels

SCE perceived variance is steeply income-graded throughout the sample (Figure 1). The weighted group means come in at Bot **25.1**, Mid 12.0, Top 9.9, so Bot’s perceived variance runs roughly **2.5** $\times$ Top’s. The gap is concentrated at the bottom: Mid and Top track each other closely and cross repeatedly, while Bot sits roughly 10 points above both for the entire

¹The cutoffs are nominal and fixed across the sample. The SCE skews toward higher incomes, so group sizes are uneven (headline sample: 1,531 / 7,501 / 11,331 observations).

sample. Bot also drifts upward over time, potentially reflecting the fixed nominal group cutoff or a greater exposure to the cycle.

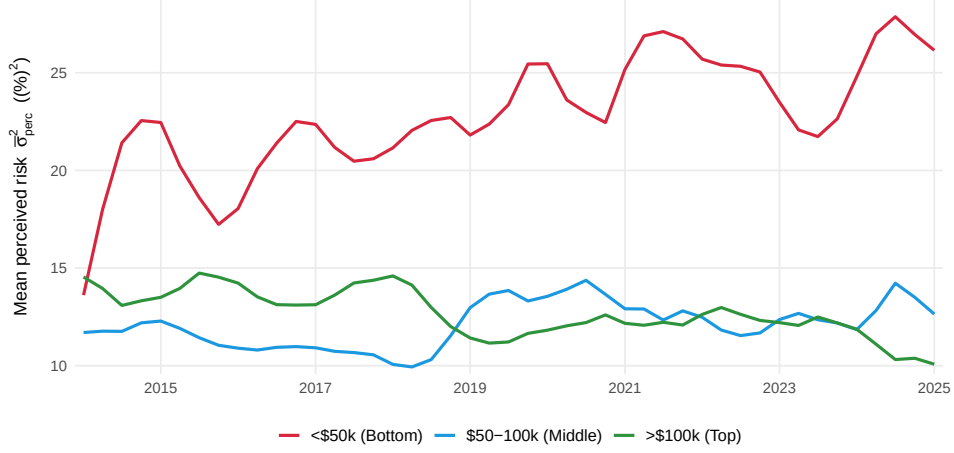


Figure 1: Group-mean perceived variance $\bar{\sigma}_{\text{perc}}^2$ in $(\%)^2$, quarterly with a 4-quarter trailing MA. Matched-stayer SCE panel, residualised as in Section 2. *Takeaway:* the Bot line sits visibly above Mid and Top across the whole sample, while Mid and Top are nearly indistinguishable.

Persistence

The level gradient says nothing about dynamics, so I next ask how persistent each group’s beliefs are. I estimate a within-respondent AR(1) on $\bar{\sigma}_{\text{perc}}^2$, with persistence coefficient $\hat{\rho}_g$, by Blundell-Bond (BB) system GMM (Blundell and Bond, 1998). The short, wide SCE panel makes the within estimator badly Nickell-biased; BB instruments the lagged level with deeper lags to remove that bias. I use BB throughout the paper following Ma et al. (2020), who work in the same short-panel-with-Nickell-bias setting:

Group	Monthly		Quarterly	
	$\hat{\rho}_g$	SE	$\hat{\rho}_g$	SE
Bot	+0.264***	0.037	+0.590***	0.050
Mid	+0.144***	0.033	+0.536***	0.077
Top	+0.146***	0.030	+0.395***	0.071
Wald ($B = T$)	$p = 0.013$		$p = 0.024$	

Table 1: Within-respondent AR(1) persistence $\hat{\rho}_g$ of perceived variance, Blundell-Bond system GMM. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 in differences plus the BB level moment, collapsed. Two-way clustered SEs (respondent, quarter). The same instrument set is used at both cadences.

Bot’s perceived variance is the most persistent of the three groups, and equality with Top is rejected at both cadences.

Monthly versus quarterly. The quarterly estimates are larger than the monthly ones for every group, which time aggregation of an AR(1) cannot deliver. The likely culprit is transitory response noise in the monthly reports, which attenuates the monthly coefficients.

The standard practice in heterogeneous-agent macro is to calibrate σ^2 from realised wage data and assume perceived equals realised. *That assumption conflicts with both the level and the persistence of perceived risk.*

4 The realised wage process cannot explain the gradient

If the belief gradient reflected a gradient in actual risk, it would show up in the realised wage process. This section shows that it does not. Realised shocks nonetheless move beliefs, and the combination pins down what a belief rule has to deliver.

Wage model

I use the standard permanent-transitory decomposition (Meghir and Pistaferri, 2004):

$$w_{i,t} = z_{i,t} + e_{i,t}, \quad e_{i,t} = p_{i,t} + \theta_{i,t}, \quad p_{i,t} = p_{i,t-1} + \psi_{i,t}.$$

Here $w_{i,t}$ is the residualised log wage, $z_{i,t}$ the predictable Mincerian path, and $e_{i,t}$ the stochastic part that the rest of the paper is about. Within e , $\psi_{i,t}$ is the permanent shock, iid with variance σ_ψ^2 , accumulating through the random walk $p_{i,t}$ (merit raise, cost-of-living adjustment); $\theta_{i,t}$ is the transitory shock, iid with variance σ_θ^2 (discretionary bonus, profit share).

Matching SCE to the model

The SCE question conditions on *same job, hours, employer*. This pins $\Delta z = 0$, so the elicited variance is the variance of the stochastic component alone:

$$\bar{\sigma}_{\text{perc},i,t}^2 \equiv \text{Var}_{i,t}(\Delta w_{i,t+1} \mid \Delta z = 0) = \text{Var}_{i,t}(\Delta e_{i,t+1}).$$

On SIPP, I residualise (removing z) and difference to recover Δe directly. The recorded wage is the paid amount per usual hour, so it contains tips, overtime, and any gap between actual and usual hours: the recorded counterpart of the paycheck the worker receives. Both sides therefore measure $\text{Var}(\Delta e)$, and the comparison is between the subjective and the realised version of the same object.

Identification of the wage primitives

The one-period change is $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$. Because ψ and θ are independent and iid, this delivers two moments per group, a variance and a first autocovariance:

$$\bar{S}_m \equiv \mathbb{E}[(\Delta e)^2] = \sigma_\psi^2 + 2\sigma_\theta^2, \quad \text{Cov}(\Delta e_t, \Delta e_{t-1}) = -\sigma_\theta^2.$$

The lag-1 autocovariance pins σ_θ^2 (only the shared θ_{t-1} term survives), and the remaining variance pins σ_ψ^2 , so the two wage primitives are exactly identified per group.

Group	σ_θ^2 (transitory)	σ_ψ^2 (permanent)	$\bar{S}_m$
Bot	2.52	11.92	16.96
Mid	4.42	11.61	20.45
Top	5.83	11.23	22.89

Table 2: SIPP permanent-transitory identification per income group, monthly cadence, in $(\%)^2$. Broad stayer panel, January seam dropped.

Permanent variance is roughly flat across groups (≈ 11.5), while transitory variance rises mildly with income, plausibly through a bonus channel at the top. The realised gradient therefore runs *opposite* to the perceived one.

The FIRE benchmark

To compare the monthly process against the 12-month SCE horizon, roll it forward 12 months under iid shocks (*derivation in Appendix B*):

$$\text{Var}(e_{t+12} - e_t) = 12\sigma_\psi^2 + 2\sigma_\theta^2, \quad (1)$$

which is what a FIRE agent who knows the parameters would report. Permanent variance scales with the horizon (one fresh innovation per month), while transitory variance enters only through the two endpoints.

Group	FIRE benchmark ($12\sigma_\psi^2 + 2\sigma_\theta^2$)	SCE observed $\bar{\sigma}_{\text{perc}}^2$	ratio
Bot	148.1	25.1	5.9×
Mid	148.2	12.0	12.4×
Top	146.4	9.9	14.8×

Table 3: FIRE-implied 12-month variance versus SCE-reported perceived variance, in $(\%)^2$.

Reading the level gap. The ratios overstate miscalibration, for a reason that is mechanical in the benchmark. Equation (1) scales permanent variance by twelve but transitory variance only by two, so any contamination that loads onto σ_ψ^2 is magnified twelvefold. SIPP earnings are self-reported, and reporting error is exactly such a contaminant. Its purely classical (iid) part differences into an MA(1) and is read as transitory, inflating σ_θ^2 ; but any *persistent* component of the error is indistinguishable from a true permanent shock, inflates σ_ψ^2 , and is then multiplied by twelve (Bound and Krueger, 1991). The benchmark is therefore biased up. Administrative payroll data tell the same story, putting job-stayers' base-wage dispersion well below the benchmark and likewise pinning the survey gap on reporting error (Grigsby et al., 2021). I therefore rest no claim on the absolute ratios. The cross-group contrast is safe: error common across groups cancels in the comparison and cannot, by itself, put opposite signs on the two gradients.

Realised shocks still move beliefs

The cross-section rules out the realised process as the *source* of the gradient, but the realised process could still drive belief *dynamics*. My hypothesis is that surprises to wage growth move perceived risk over time. Figure 2 supports this: pooled perceived risk co-moves with the absolute deviation of aggregate wage growth from its sample mean.²

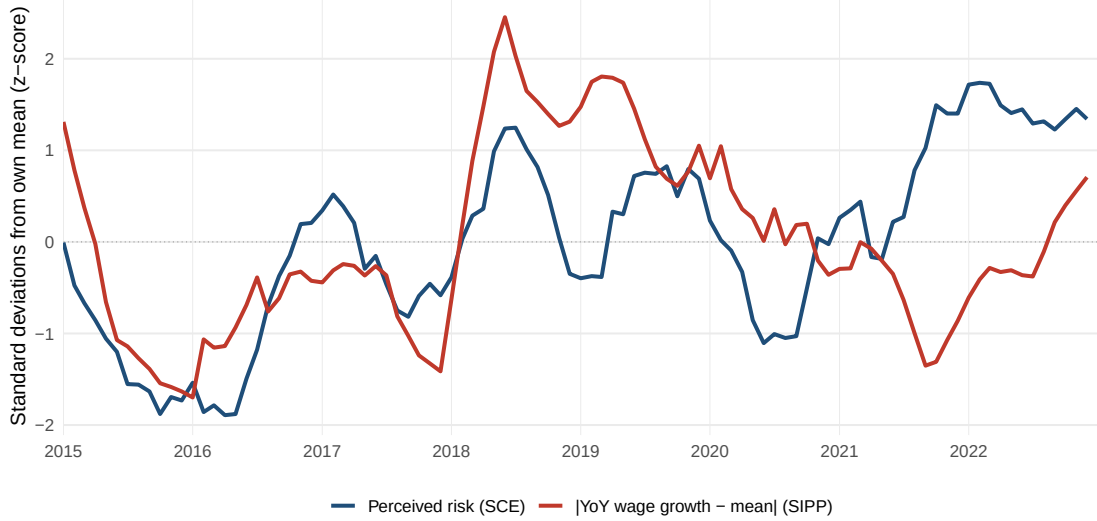


Figure 2: SCE perceived risk and the absolute deviation of SIPP aggregate wage growth from its sample mean, both standardised, pooled across income groups, 2015–2022. *Takeaway:* the two series move together, so realised wage news shifts perceived risk.

Two requirements for a belief rule. Together, these facts impose two requirements on any belief rule. First, it must let perceived variance move, since Figure 1 rules out constant beliefs. Second, it must let realised shocks drive that movement (Figure 2); and since the realised process is mostly common across groups, the rule must let the response to that shared signal differ by group.

5 A belief rule that nests FIRE

The rule I take to the data is the simplest one that meets both requirements.

The agent's problem

The agent cannot see permanent and transitory shocks separately. Each period she observes one squared pay change, which mixes the two ($\mathbb{E}[(\Delta e)^2] = \sigma_\psi^2 + 2\sigma_\theta^2$), and must attribute the surprise: a share, which I will call the attribution share λ , to permanent earning power, and the rest to a transitory blip. The rule is Bayesian throughout: the agent is not assumed to over- or under-react, and her deviations from FIRE come entirely from what she can observe. Section 8 microfound this observation structure.

²Absolute deviations, since any realised shock should raise perceived uncertainty; squared deviations give a very similar plot. The motivating series is year-on-year aggregate growth; the estimation regressor in Section 6 is the monthly cell-mean squared change. Section 7 splits shocks by sign.

Modelling choice

The SCE delivers one perceived-variance series per group, which is enough to pin one dynamic belief, not two. I therefore let the permanent belief $\hat{\sigma}_{\psi,t}^2$ evolve and treat the transitory belief as a static prior, $\sigma_{\theta,\text{perc}}^2$, constant per group. The permanent component is the welfare-relevant one (permanent shocks shift lifetime earning power; transitory ones fade quickly), so this is where the dynamics should live. The cost of the asymmetry is that any unmodelled belief-side mis-attribution gets absorbed into the attribution share λ introduced next.

The updating rule

The permanent-variance belief follows an exponentially weighted moving average (EWMA) of past squared shocks, with the attribution share entering as a separate weight:

$$\hat{\sigma}_{\psi,t}^2 = \underbrace{(1-\mu)}_{\text{persistence}} \hat{\sigma}_{\psi,t-1}^2 + \underbrace{\mu}_{\text{update weight}} \underbrace{\lambda}_{\text{attribution share}} (\Delta e_{t-1})^2. \quad (2)$$

The two parameters are:

- μ , the update speed. Higher μ means each period's news moves the belief more, and $1-\mu$ is the carryover of last period's belief.
- λ , the share of each squared surprise treated as informative about the permanent component. $\lambda = 0$ means every shock is read as fully transitory.

Because the two weights $(1-\mu)$ and μ sum to one, equation (2) is the integrated GARCH(1,1) recursion in the squared surprise, which is also the steady-state Kalman filter that tracks an unknown permanent variance modelled as a random walk: μ plays the role of the steady-state Kalman gain. Appendix F gives the filter form and shows how that gain is pinned by the signal-to-noise ratio.

FIRE inside the rule

Substituting the belief into the FIRE identity (1) gives the level the agent reports each period:

$$\bar{\sigma}_{\text{perc},t}^2 = 12 \hat{\sigma}_{\psi,t}^2 + 2 \sigma_{\theta,\text{perc}}^2. \quad (3)$$

Strict FIRE is the corner $\mu = 0$ with both beliefs at the truth: reported variance is then constant at $12\sigma_{\psi}^2 + 2\sigma_{\theta}^2$, a corner the dynamics of Section 3 rule out. Within the learning family ($\mu > 0$), FIRE survives only as a long-run benchmark. Take unconditional expectations of (2) in a stationary state: the persistence terms cancel ($\mathbb{E}[\hat{\sigma}_{\psi,t}^2] = \mathbb{E}[\hat{\sigma}_{\psi,t-1}^2]$), leaving $\mathbb{E}[\hat{\sigma}_{\psi}^2] = \lambda \mathbb{E}[(\Delta e)^2] = \lambda \bar{S}_m$. The belief is therefore correct on average exactly when

$$\lambda^* = \sigma_{\psi}^2 / \bar{S}_m,$$

which Table 2 puts at 0.70/0.57/0.49 for Bot/Mid/Top. The transitory side adds its own condition, $\sigma_{\theta,\text{perc}}^2 = \sigma_{\theta}^2$. Section 7 shows neither condition holds.

6 Estimation

The belief recursion (2) is in the unobserved $\hat{\sigma}_\psi^2$. Combining it with the level identity (3) substitutes the belief out and leaves an AR(1) in the *observable* perceived variance (*derivation in Appendix A*):

$$\bar{\sigma}_{\text{perc},i,t}^2 = \underbrace{2\mu_g \sigma_{\theta,\text{perc},g}^2}_{\beta_{\text{cons}}} + \underbrace{(1 - \mu_g)}_{\beta_{\text{lag}}} \bar{\sigma}_{\text{perc},i,t-1}^2 + \underbrace{12\mu_g \lambda_g}_{\beta_{\text{shock}}} (\Delta e_{i,t-1})^2 + \alpha_i + u_{i,t}, \quad (4)$$

where α_i is a person fixed effect and $u_{i,t}$ an idiosyncratic shock. The three reduced-form coefficients map invertibly to the three structural parameters:

$$\mu_g = 1 - \hat{\beta}_{\text{lag}}, \quad \sigma_{\theta,\text{perc},g}^2 = \frac{\hat{\beta}_{\text{cons}}}{2\mu_g}, \quad \lambda_g = \frac{\hat{\beta}_{\text{shock}}}{12\mu_g}.$$

Each coefficient is identified off a distinct feature of the data: β_{lag} off the within-respondent persistence of perceived variance, β_{cons} off the group level once the news loading is netted out, and β_{shock} off how much perceived variance moves with the lagged realised shock.

The shock regressor

The rule (2) calls for the agent's own $(\Delta e_{i,t-1})^2$. The SCE has no own wage histories, so I substitute the SIPP cell mean over workers in the same cell $c = (\text{group} \times \text{Census division} \times \text{month})$, following the JMP of Guerreiro (2023). Workers in the same local labour market share components of wage news, absorbed through coworkers and local media.³ Census divisions balance statistical power against geographic resolution. The cell series is smoothed with a 3-month trailing MA.

³On the salience of personal and local experience for expectations, see e.g. Kuchler and Zafar (2019).

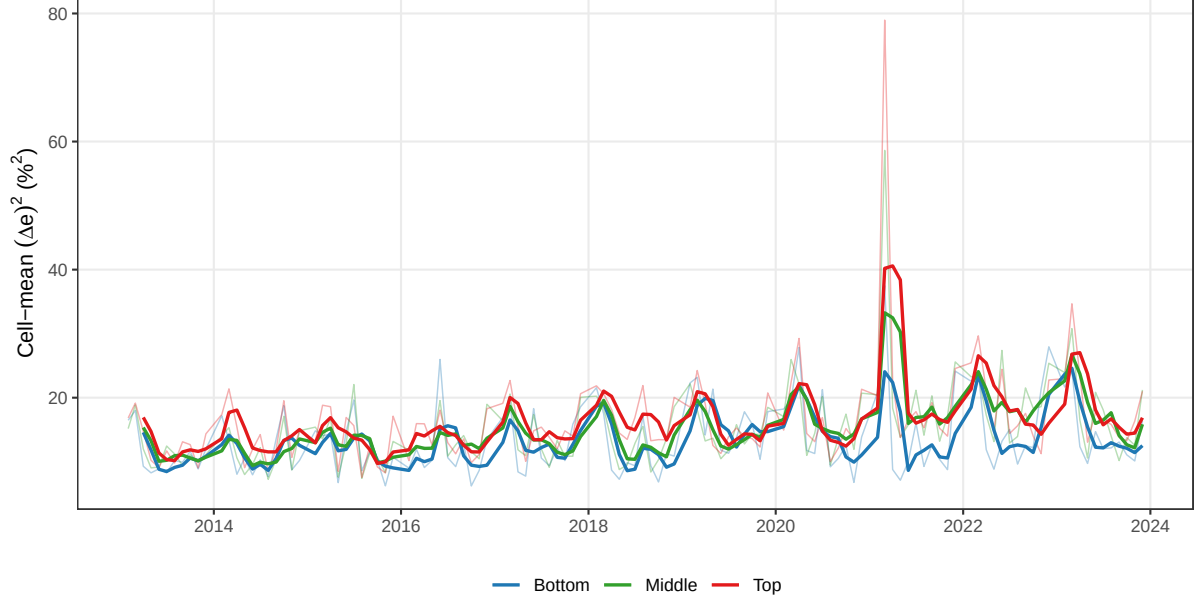


Figure 3: Cell-mean $(\Delta e_{i,t})^2$ by income group, national monthly, in $(\%)^2$. Thin lines: raw monthly cell means. Thick lines: the 3-month trailing MA used as the regressor in (4). *Takeaway:* Top has the largest shock amplitude, and Bot’s is no larger, so any stronger belief response at Bot cannot come from larger shocks.

Top shows the largest shock amplitude, consistent with its higher σ_θ^2 in Table 2 and with discretionary pay concentrating at the top (Güvenen et al., 2021; Halvorsen et al., 2024). Since Bot’s shocks are no larger than Top’s, any stronger belief response at Bot is a property of beliefs, not of shocks.

Identification and caveats

Estimand. The rule calls for the agent’s own squared surprise, but the regressor is a cell mean. Replacing the own shock with a noisy group proxy is textbook errors-in-variables, and it attenuates the loading: β_{shock} , and hence $\hat{\lambda}$, is biased toward zero. Cell averaging removes idiosyncratic variation identically in every group, so the attenuation factor is roughly common across groups. The *level* of $\hat{\lambda}$ is therefore not interpretable on its own, but cross-group *ratios* divide the common factor out. I read $\hat{\lambda}$ in ratios, not levels.

Instruments. I instrument with $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 and the shock at lags 1–3, plus BB level moments. The shock enters as endogenous: the contemporaneous shock and the belief share a survey-timing and aggregate-news channel, and consistent with this the Hansen J test of the over-identifying restrictions rejects the contemporaneous moment set while accepting lags 1–3.

7 Headline: the bottom updates slowest, responds most, and holds an inflated prior

Joint GMM on (4) by income group delivers the estimates in Table 4. The three columns of structural parameters are read against the SIPP truth in the first column.

Group	σ_θ^2 (SIPP truth)	$\sigma_{\theta,\text{perc},g}^2$ (GMM)	λ_g (GMM)	μ_g (GMM)	Hansen J p	N
Bot	2.52	9.75 (1.71)	+0.077 (0.030)	0.628 (0.055)	0.20	1,531
Mid	4.42	2.94 (0.93)	+0.046 (0.015)	0.912 (0.089)	0.23	7,501
Top	5.83	4.12 (0.82)	+0.026 (0.013)	0.950 (0.060)	0.24	11,331
Wald ($B = T$)	—	$p = \mathbf{0.003}$	$p = 0.12$	$p = \mathbf{0.0001}$	—	—

Table 4: Joint Blundell-Bond system GMM on (4), by income group. Two-way clustered SEs (respondent, quarter) in parentheses. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4, the shock at lags 1–3, plus BB level moments. The SIPP truth column repeats Table 2; the Hansen J never rejects.

Attribution share λ

All three groups read part of each wage shock as a signal about permanent earning power: each $\hat{\lambda}_g$ is significant and positive. Point estimates fall monotonically in income, with Bot’s three times Top’s, but cross-group equality is not rejected ($p = 0.12$). *Implication.* A common wage signal generates larger revisions of the permanent-variance belief at the bottom, in point estimates if not yet in significance.

Transitory prior $\sigma_{\theta,\text{perc}}^2$

Realised transitory variance rises with income; the perceived prior does the opposite. Bot’s prior sits almost four times above her realised value, while Mid and Top sit near or below theirs. The bottom is the only group whose prior is materially detached from its realised data; Section 8 supplies the reading-noise channel that opens the gap. The gradient is significant ($p = 0.003$). *Implication.* Bot’s inflated prior is the dominant driver of the *level* of her perceived variance, separate from the news channel.

Update speed μ

Bot updates significantly more slowly than Mid or Top. The implied persistence ordering matches Table 1 (Bot most persistent); the levels probably differ because the shock regressor absorbs part of the autocorrelation and the monthly estimates are noise-attenuated (Section 3). *Implication.* Bot’s perceived risk embeds older information, so any wage news lives longer in her beliefs.

Robustness

Occupation. Adding an occupation $\times$ month fixed effect to the shock construction preserves the persistence and attribution gradients and sharpens the latter; the prior gradient weakens (Appendix D.1). I do not adopt it as the headline: low-income workers sort into occupations with structurally higher uncertainty (Morduch and Schneider, 2017), so partialling out occupation removes part of the channel itself, not just confounding variation.

Sweeps. Appendix D.2 varies instrument lags, the trim, and COVID handling one dial at a time. The persistence gradient survives every variant except the tighter within-group trim, and the attribution gradient stays monotone throughout, reaching significance in one variant. The

prior gradient loses significance without 2020, so part of Bot’s elevated prior is identified off the COVID year.

Asymmetry. Splitting the shock regressor by sign and pooling across groups for power (Appendix D.3), both components load positively, with positive news carrying roughly twice the weight of negative (0.055 vs 0.027). The direction is the opposite of what loss aversion would predict and I read it as: upside news is read as informative about future earning power, while downside news is more often dismissed as transitory.

8 Where the wedges come from: pay delivery, not pay risk

The headline leaves the three parameters per group structurally unconnected. I now link all three to two primitives, both observable features of how pay is delivered.

Two microfounded primitives

Noisy reading of own pay. This channel is standard in the literature: the worker does not read her wage process off a clean statement. Pay at the bottom arrives as a stream of fragments: variable shift counts, cash tips, irregular supplements. For instance, a bartender reconstructing March’s wage change aggregates many separate shift payouts, whereas a salaried manager reads one number. I model her real-time reading of the month’s pay as

$$\tilde{y}_{i,t} = e_{i,t} + \nu_{i,t}, \quad \nu_{i,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \omega_g^2), \quad (5)$$

a signal-extraction structure in the style of Pischke (1995), where $\tilde{y}_{i,t}$ is the perceived paycheck and $\nu_{i,t}$ is *perception* noise of variance ω_g^2 , rather than income variation.⁴ She feeds $(\Delta \tilde{y}_t)^2$ into the belief rule (2) as if it were $(\Delta e_t)^2$.

Labelled transitory pay. High-income compensation includes visibly transitory components: annual bonuses and sales commissions arrive as separate line items (Lemieux et al., 2009), and high-income shocks are indeed predominantly transitory (Güvenen et al., 2021). By contrast, low-income compensation is hours times wage plus tips, with no flag separating the transitory portion. The same logic covers macro exposure: the top’s tends to arrive in classifiable form (furloughs with return dates, cyclical hours cuts), while the bottom’s arrives in more ambiguous form (sector closures, gig disappearance) (Hoynes et al., 2012). Let $\rho_g \in [0, 1]$ be the fraction of transitory pay that arrives institutionally labelled as transitory, with $\rho_{\text{Top}} \gg \rho_{\text{Bot}}$. The labelled share is netted out before the belief update; only the unlabelled residual feeds the belief rule (2), where she still has to attribute it between permanent and unlabelled transitory.

Both primitives are income-graded by construction. Both gradients are what rational inattention predicts (Sims, 2003): scarce attention is spent where it pays most. It pays least at the bottom, where pay is harder to read and the consumption stakes per unit of precision are

⁴Recorded SIPP earnings capture the amounts actually paid, so ν does not enter the moments of Section 4. Monthly income at the bottom is dominated by hours, gig, and tip flows (Hardy and Ziliak, 2014; Morduch and Schneider, 2017); the claim here is that such fragmented delivery makes the worker’s own reading of her pay noisy.

lower, so ω_g^2 rises. For the same reason ρ_g falls, as less attention is spent parsing low-income pay into labelled and unlabelled parts.

From primitives to estimated gradients

The two primitives map onto the three estimated parameters: the labelled share ρ_g governs how fast beliefs update and sets the attribution slope, while paycheck noise ω_g^2 lifts the transitory prior (which labelling pulls back down).

Update speed. The gain is set by how cleanly pay separates into permanent and transitory parts. More labelling shrinks the unlabelled residual variance $V_u = \sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2$ relative to the total $S = \sigma_\psi^2 + 2\sigma_\theta^2$, so each squared surprise is a sharper signal about permanent risk and the steady-state Kalman gain is larger. The top, labelling most, updates fastest; the bottom, labelling nothing, filters the noisiest unlabelled mix and updates slowest. (Reading noise washes out of the gain: it scales the squared signal and its noise together, leaving the signal-to-noise ratio unchanged, so it acts on the prior, not the speed.) This is the μ column of Table 4, equivalently the persistence gradient of Table 1 (*filter form and algebra in Appendix F*).

Transitory prior. Here the two primitives pull in opposite directions. Reading noise pushes the prior up. Differencing the noisy reading (5) leaves the noise as $\nu_t - \nu_{t-1}$, an MA(1) that carries the same variance-and-autocovariance signature as a true transitory shock.⁵ Because transitory variance is read off exactly that autocovariance (Section 4), the agent cannot tell the two apart and books ω_g^2 as extra transitory risk. Labelling pushes the other way: the flagged share is identified and netted out before she updates, leaving only the unlabelled $(1 - \rho_g)\sigma_\theta^2$ as risk she still carries. The prior is the sum of the two,

$$\sigma_{\theta,\text{perc},g}^2 = (1 - \rho_g)\sigma_\theta^2 + \omega_g^2, \quad (6)$$

and which force wins flips across the distribution. At the bottom noise dominates (high ω^2 , low ρ) and the prior sits well above realised transitory variance; higher up labelling dominates (low ω^2 , high ρ) and it sits below. So while realised transitory variance rises with income (Table 2), perceived does not: Bot steps far above its truth and Mid and Top fall below theirs, the bidirectional (U-shaped, even) pattern of Table 4.

Attribution share. The last parameter is a slope, and labelling flattens it. The agent updates her permanent belief only on the unlabelled part of a shock; the labelled part is already known to be transitory and says nothing about permanent risk. The regression, though, has only the full shock to load on, labelled and unlabelled together. The labelled part therefore adds movement to the regressor with no matching movement in the belief, exactly like noise added to a regressor, and attenuates the slope: the larger the labelled share of the shock's variance, the flatter the fitted λ_g . Under joint Gaussianity of the wage shocks ψ and θ (*derivation in*

⁵Both differences are MA(1) of the same shape: $\theta_t - \theta_{t-1}$ has variance $2\sigma_\theta^2$ and first autocovariance $-\sigma_\theta^2$, and $\nu_t - \nu_{t-1}$ has variance $2\omega_g^2$ and first autocovariance $-\omega_g^2$. The permanent shock is serially uncorrelated and drops out of the autocovariance, so a transitory innovation of variance ω_g^2 is its exact observational twin.

Appendix C),

$$\lambda_g = \frac{\sigma_\psi^2(\sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2}, \quad (7)$$

which decreases in ρ_g and equals the FIRE benchmark λ^* when nothing is labelled ($\rho_g = 0$). On the labelling channel Bot, labelling least, sits closest to that benchmark and Top furthest from it. Equation (7) abstracts from reading noise, which scales the slope down by a factor below one that is largest where the noise is largest, namely at the bottom (the remark to Appendix C); the monotone λ column of Table 4 is the net of the two forces. The cell-mean proxy shifts all three down by roughly the same factor (Section 6), so I read only their ratios.

No bias required

Given (ω_g^2, ρ_g) , the agent neither over- nor under-reacts to her signal relative to the Bayesian benchmark; her deviations from FIRE come from what she can observe, not from how she reacts to it. The income gradient in beliefs is a gradient in information technology, not in rationality. This separates the mechanism from over-extrapolation accounts such as diagnostic expectations, which predict over-reaction relative to the Bayesian benchmark. Since the labelled share is observable as ρ_g , we incorporate the superior-information critique of (Kaufmann and Pistaferri, 2009): a perceived-versus-econometric gap can reflect what the worker knows and the data cannot see, and here that knowledge is just ρ_g .

9 Embedding in a heterogeneous-agent model

Setup

I embed the estimated belief structure in an Aiyagari-style buffer-stock economy, solved in partial equilibrium (general equilibrium is future work; see Section 10). Liquidity is group-specific and rises steeply with income: the bottom is near hand-to-mouth, the top holds a multi-month buffer. This gradient alone reproduces the steep empirical fall of MPCs with income; the belief wedge then sits on top of it.

Income follows the permanent-transitory process of Section 4, $y_{i,t} = P_{i,t} \exp(\theta_{i,t})$ with $P_{i,t} = P_{i,t-1} \exp(\psi_{i,t})$ and group-specific variances from Table 2 (permanent ≈ 11.5 throughout, transitory rising from 2.5 to 5.8). Following Carroll (2006), I normalise by permanent income: the state is cash-on-hand per unit of permanent income, the per-period income draw is the transitory $\exp(\theta_g)$, and permanent shocks enter through the growth factor $\exp(\psi_g)$. The problem is scale-invariant, so I report each group’s mean monthly SCE income (\$2,415 / \$6,014 / \$13,604) only to fix magnitudes.⁶

Two regimes. I solve each group twice. Under the *belief* regime the agent perceives wage risk through the estimated rule of Section 7 and her belief evolves; under *FIRE* she knows the true variances, so beliefs sit at the realised process. Everything else is shared, so any gap between the two is the belief wedge. FIRE here means correct *wage-risk* beliefs; job-loss risk is common

⁶The wage variances are stated in $(\%)^2$ throughout; in the income process they enter in fractional units (divided by 10^4) before exponentiating, and the perceived-variance state is carried back in $(\%)^2$.

to both and nets out.

Unemployment. The SCE measures *same-job* risk and is silent on job loss, so the budget adds a standard two-state employment block: monthly separation 0.034 and job-finding 0.45 (Shimer, 2005), with replacement income half of permanent, implying 7 percent unemployment. Common across groups and regimes, it lifts the level of precaution but adds nothing to the cross-group or belief-versus-FIRE comparison. Beliefs update only while employed and freeze during unemployment, when no wage signal arrives.

The household problem

$$V_g(a, y, \hat{\sigma}_\psi^2) = \max_c \left\{ u(c) + \beta \mathbb{E}[V_g(a', y', \hat{\sigma}_{\psi'}^2) \mid y, \hat{\sigma}_\psi^2] \right\}, \quad \text{s.t. } a' = Ra + y - c, \quad a' \geq \underline{a}.$$

The states are assets a , income y , and the permanent-variance belief $\hat{\sigma}_\psi^2$ ($R = 1 + r$, borrowing limit $\underline{a} = 0$); it is written here in levels but solved in the permanent-income-normalised form introduced above. The two information channels enter separately: the budget uses *realised* income, while expectations are taken under the agent's *perceived* process, so beliefs act only through the saving motive, and a higher $\hat{\sigma}_\psi^2$ raises perceived risk and lifts saving. The agent observes her permanent income (she normalises by it), so beliefs distort the perceived *variance* of future shocks, not the attribution of the current one. The model therefore isolates this second-moment channel and leaves the first-moment signal-extraction of Pischke (1995) aside. The belief follows the learning rule of Section 5: she squares her reading of the month's pay change and updates by the gain μ_g times the attribution share λ_g ,

$$\hat{\sigma}_{\psi,t}^2 = (1 - \mu_g) \hat{\sigma}_{\psi,t-1}^2 + \mu_g \lambda_g (\Delta \tilde{y}_t)^2.$$

Both objects are the steady-state primitives of Appendix F: μ_g the gain that tracks the permanent variance through the squared signal, λ_g the labelled-pay attribution. Because λ_g scales the signal, the belief settles around $\lambda_g \mathbb{E}[(\Delta \tilde{y})^2]$, which the labelling discount holds below the realised permanent variance at the middle and top: they perceive less permanent risk than they face, the second-moment counterpart of the level gap of Section 4. A realised shock drives the next pay reading, which updates the belief, closing the loop between income and beliefs.

Calibration

Calibrating the model fixes two things per group: how patient it is (the discount factor β_g) and what it believes about wage risk (the gain μ_g , the attribution λ_g , and the transitory prior $\sigma_{\theta, \text{perc}, g}^2$). I set them in two steps.

Preferences, to match liquid wealth. Preferences are CRRA ($\gamma = 2$, $r = 2$ percent annual). For each group I bisect β_g until its FIRE economy holds that income group's liquid wealth in the data, about 0.5/2.0/3.0 months (achieving 0.6/2.0/3.0; the bottom binds at an impatience floor), giving $\beta = 0.80/0.93/0.95$.⁷ The bottom is impatient and near the borrowing constraint,

⁷Targets from the SCE Household Finance module: non-retirement savings and investments (the liquid-wealth concept of Kaplan et al., 2014) relative to income. Counting the households that report none as zero (40 percent of the bottom income group, against 13 percent of the top), median liquid wealth is about 0.2/1.5/3.2 months

the top patient and cushioned. I solve by the endogenous grid method (Carroll, 2006) and simulate each regime.

Beliefs, from the two pay-delivery primitives. Rather than impose the belief parameters, I let them follow from the mechanism of Section 8. I fit the reading noise ω_g^2 and the labelled share ρ_g , with a common drift q , to the estimated belief moments; the gain, attribution, and prior then follow from the steady-state filter (Appendix F), the attribution identity (7), and the prior identity (6). What the household block uses is not an absolute attribution level: the perceived permanent variance settles at the realised σ_ψ^2 scaled by a labelling discount, $\sigma_\psi^2 (V_u/S)^2$ with $V_u = \sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2$ and $S = \sigma_\psi^2 + 2\sigma_\theta^2$, built from the SIPP variances and the labelled share ρ_g alone. Only the attribution ratios are targeted in the fit (Section 6); they pin ρ_g , never λ 's level. Bot, labelling nothing, perceives the realised variance; the middle and top a fraction of it. Two primitives per group and one common drift, seven numbers for eight moments, reproduce all three gradients (Table 5): the gains, the top's included; the prior gradient; and the attribution gradient, matched in ratios since the cell-mean proxy attenuates its level (Section 6). The fit is not rejected ($J = 0.1$). The primitives are economically ordered: reading noise is highest at the bottom and labelling rises steeply with income, so the bottom reads its pay through the most noise and has the least flagged as transitory. Identification leans on the extremes, driving the labelled share to its bounds, none of the bottom's transitory pay flagged and all of the top's.

Group	Gain μ_g		Prior $\sigma_{\theta,\text{perc},g}^2$		Primitives	
	data	fit	data	fit	ρ_g	ω_g^2
Bot	0.628	0.628	9.75	9.51	0.00	6.99
Mid	0.912	0.904	2.94	2.93	0.92	2.58
Top	0.950	0.956	4.12	4.22	1.00	4.22

Table 5: Belief moments generated by the two pay-delivery primitives. The reading noise ω_g^2 and labelled share ρ_g are fit per group with a common drift q ; the gain, prior, and attribution follow from Appendix F, (7), and (6). Gain and prior matched in levels, attribution in ratios (Bot/Top, Mid/Top): data 2.96/1.77, fit 2.80/1.75. Over-identification $J = 0.1$ (1 df).

Steady-state results

Table 6 sets the belief regime against an otherwise identical FIRE benchmark that holds beliefs at the truth.

across the three groups, a steep rise with income; I round to 0.5/2.0/3.0, the top capped by the GIC ceiling and the bottom by the impatience floor. There is no mobility across groups, a short-horizon device: over longer spans permanent shocks would themselves carry households across group lines.

Group	Mean wealth (months)			12-month MPC		
	FIRE	Belief	Δ	FIRE	Belief	Δ
Bot	0.60	0.60	-1%	0.72	0.72	+0.2pp
Mid	2.01	1.60	-20%	0.34	0.41	+7.1pp
Top	3.02	1.98	-34%	0.22	0.34	+11.8pp

Table 6: Belief regime versus FIRE, by income group, under group-specific liquidity. Wealth in months of permanent income; MPC over 12 months out of a small transitory windfall. PE, $r = 2$ percent, $\beta = 0.80/0.93/0.95$.

Beliefs barely move the bottom (within a percent on wealth, 0.2 points on the MPC): hand-to-mouth and constrained, and perceiving permanent risk close to the truth (λ_{Bot} is unbiased at $\rho = 0$), it has neither the room nor the wedge to act on. The middle and top respond sharply, holding 20 and 34 percent less wealth and consuming 7 and 12 points more of a windfall. They perceive far less permanent risk than they face: the belief settles at $\sigma_\psi^2 (V_u/S)^2$, here 4.1 and 2.7 against a realised 11.6 and 11.2, and they perceive less transitory risk too (the labelling discount), so both moments push them to save less. The wedge therefore falls on the middle and top, not the bottom, whose perceived risk is most distorted but whose behaviour liquidity sets. It now runs through both moments: wherever pay is labelled the permanent-variance belief sits below the truth, so the wedge bites in the steady state and not only after a shock.

10 Dynamics: the response to a recession

The steady-state comparison was a level statement. The dynamic counterpart asks how the belief wedge shapes the response to an aggregate shock. I hit the stationary economy with a single negative innovation to the common permanent wage component, a permanent 3 percent level fall, which cleanly separates the shared income loss from the belief response. It lowers income and, through the squared signal, raises perceived risk. I trace each regime against its own no-shock path, then plot the belief-regime deviation minus the FIRE deviation (Figure 4).

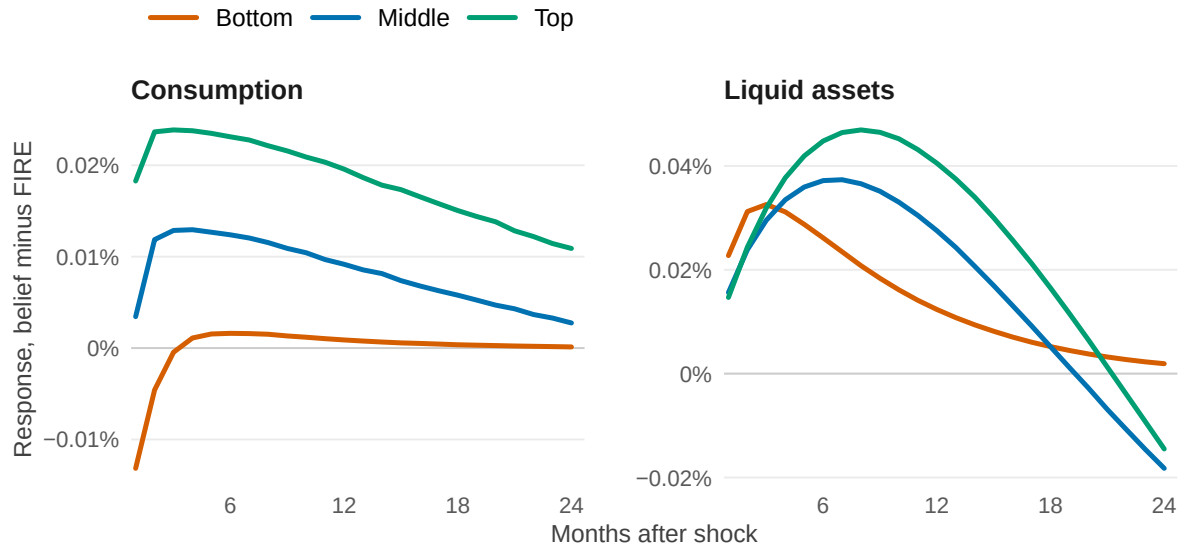


Figure 4: Belief channel (belief regime minus FIRE) after the recession shock: consumption (left) and end-of-period assets (right), in percent of the no-shock baseline, by income group.

The belief channel is small, and smaller than the steady-state wedge. Consumption falls about 2.5 percent on impact in both regimes, the shared permanent-income loss; perceived risk rises 13–23 (%)², so the belief regime adds at most a few hundredths of a percent to the consumption response, fading within a quarter for the middle and top, more slowly at the bottom (its lower gain $\mu = 0.63$). The multiplicative rule scales each revision by the attribution share, so the dynamic response is muted even where the steady-state wedge (Table 6) is large: the action is in the chronic level of perceived risk, not in its reaction to a single shock. Liquidity barely matters here, unlike the steady state: a recession raises perceived risk, so the belief response is extra precautionary saving, which the borrowing constraint never blocks (it limits dissaving, not saving), whereas the steady-state wedge lowers desired buffers, which the constraint can bind against. The cross-group differences therefore appear in the steady-state cross-section, not the dynamics. The monthly gain overstates the volatility of perceived risk, so a quarterly gain would lengthen its persistence; the small dynamic channel is therefore a monthly-cadence statement.

Interest rates. The model is partial equilibrium, so r is fixed. In general equilibrium the belief wedges would move it by shifting aggregate saving, which is why the model carries the full Aiyagari wealth distribution; the sign is unclear, since the bottom over-saves but the middle and top hold most of the wealth and save less. This is my next extension I want to work on.

References

- Blundell, R. and Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87(1):115–143.
- Bound, J. and Krueger, A. B. (1991). The extent of measurement error in longitudinal earnings data: Do two wrongs make a right? *Journal of Labor Economics*, 9(1):1–24.
- Broer, T., Kohlhas, A., Mitman, K., and Schlafmann, K. (2022). Expectation and wealth heterogeneity in the macroeconomy. Working paper.
- Caplin, A., Gregory, V., Lee, E., Leth-Petersen, S., and Sæverud, J. (2023). Subjective earnings risk. NBER Working Paper 31019, National Bureau of Economic Research.
- Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320.
- Grigsby, J., Hurst, E., and Yildirmaz, A. (2021). Aggregate nominal wage adjustments: New evidence from administrative payroll data. *American Economic Review*, 111(2):428–471.
- Guerreiro, J. (2023). Belief disagreement and business cycles. Job Market Paper, Northwestern University.
- Güvenen, F., Karahan, F., Ozkan, S., and Song, J. (2021). What do data on millions of US workers reveal about lifecycle earnings dynamics? *Quarterly Journal of Economics*, 136(4):2349–2391.
- Halvorsen, E., Holter, H. A., Ozkan, S., and Storesletten, K. (2024). Dissecting idiosyncratic earnings risk. *Review of Economic Studies*.
- Hardy, B. and Ziliak, J. P. (2014). Decomposing trends in income volatility: The “Wild Ride” at the top and bottom. *Economic Inquiry*, 52(1):459–476.
- Hoynes, H., Miller, D. L., and Schaller, J. (2012). Who suffers during recessions? *Journal of Economic Perspectives*, 26(3):27–48.
- Kaplan, G., Violante, G. L., and Weidner, J. (2014). The wealthy hand-to-mouth. *Brookings Papers on Economic Activity*, pages 77–138.
- Kaufmann, K. and Pistaferri, L. (2009). Disentangling insurance and information in intertemporal consumption choices. *American Economic Review: Papers & Proceedings*, 99(2):387–392.
- Kim, G. and Binder, C. (2023). Learning-through-survey in inflation expectations. *American Economic Journal: Macroeconomics*, 15(2):254–278.
- Kuchler, T. and Zafar, B. (2019). Personal experiences and expectations about aggregate outcomes. *Journal of Finance*, 74(5):2491–2542.
- Lemieux, T., MacLeod, W. B., and Parent, D. (2009). Performance pay and wage inequality. *Quarterly Journal of Economics*, 124(1):1–49.
- Ma, Y., Ropele, T., Sraer, D., and Thesmar, D. (2020). A quantitative analysis of distortions in

- managerial forecasts. NBER Working Paper 26830, National Bureau of Economic Research.
- Meghir, C. and Pistaferri, L. (2004). Income variance dynamics and heterogeneity. *Econometrica*, 72(1):1–32.
- Morduch, J. and Schneider, R. (2017). *The Financial Diaries: How American Families Cope in a World of Uncertainty*. Princeton University Press.
- Pischke, J.-S. (1995). Individual income, incomplete information, and aggregate consumption. *Econometrica*, 63(4):805–840.
- Rozsypal, F. and Schlafmann, K. (2023). Overpersistence bias in individual income expectations and its aggregate implications. *American Economic Journal: Macroeconomics*, 15(4):331–371.
- Shimer, R. (2005). The cyclical behavior of equilibrium unemployment and vacancies. *American Economic Review*, 95(1):25–49.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690.
- Wang, T. (2023). Perceived versus calibrated income risks in heterogeneous-agent consumption models. Bank of Canada Staff Working Paper 23-59.

A From EWMA to observable AR(1)

Two ingredients: the EWMA (2) and the level identity (3).

Step 1. Solve (3) for the belief:

$$\hat{\sigma}_{\psi,t}^2 = \frac{\bar{\sigma}_{\text{perc},t}^2 - 2\sigma_{\theta,\text{perc}}^2}{12}.$$

Step 2. Substitute both $\hat{\sigma}_{\psi}^2$ terms into (2) and multiply by 12:

$$\bar{\sigma}_{\text{perc},t}^2 - 2\sigma_{\theta,\text{perc}}^2 = (1 - \mu) [\bar{\sigma}_{\text{perc},t-1}^2 - 2\sigma_{\theta,\text{perc}}^2] + 12\mu\lambda (\Delta e_{t-1})^2.$$

Step 3. Collect the intercept: $2\sigma_{\theta,\text{perc}}^2 - 2(1 - \mu)\sigma_{\theta,\text{perc}}^2 = 2\mu\sigma_{\theta,\text{perc}}^2$.

Step 4. Add group subscripts, a person fixed effect α_i , and an idiosyncratic shock $u_{i,t}$:

$$\bar{\sigma}_{\text{perc},i,t}^2 = 2\mu_g \sigma_{\theta,\text{perc},g}^2 + (1 - \mu_g) \bar{\sigma}_{\text{perc},i,t-1}^2 + 12\mu_g \lambda_g (\Delta e_{i,t-1})^2 + \alpha_i + u_{i,t}.$$

This is (4).

B 12-month FIRE variance

Roll the monthly process $e_t = p_t + \theta_t$ forward H months.

Step 1. Split the H -period change:

$$e_{t+H} - e_t = \underbrace{(p_{t+H} - p_t)}_{\text{permanent}} + \underbrace{(\theta_{t+H} - \theta_t)}_{\text{transitory}}.$$

Step 2. Permanent piece. The random walk accumulates one fresh innovation per month:

$$p_{t+H} - p_t = \sum_{k=1}^H \psi_{t+k}, \quad \text{Var}(p_{t+H} - p_t) = H\sigma_{\psi}^2.$$

Step 3. Transitory piece. θ is iid, so only the endpoints matter:

$$\text{Var}(\theta_{t+H} - \theta_t) = 2\sigma_{\theta}^2.$$

Step 4. The shock processes are independent, so the variances add:

$$\text{Var}(e_{t+H} - e_t) = H\sigma_{\psi}^2 + 2\sigma_{\theta}^2.$$

At $H = 12$: $12\sigma_{\psi}^2 + 2\sigma_{\theta}^2$, which is (1).

C Closed form for λ_g under partial labelling

Setup. Split the transitory shock $\theta_t = \theta_t^{\text{lab}} + \theta_t^{\text{unlab}}$ into independent labelled and unlabelled pieces with variances $\rho_g \sigma_\theta^2$ and $(1 - \rho_g) \sigma_\theta^2$. Substituting into $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$ and grouping by labelling status:

$$\Delta e_t = \underbrace{\psi_t + \theta_t^{\text{unlab}} - \theta_{t-1}^{\text{unlab}}}_{u_t, \text{ unlabelled residual}} + \underbrace{\theta_t^{\text{lab}} - \theta_{t-1}^{\text{lab}}}_{l_t, \text{ labelled component}},$$

with $V_u \equiv \text{Var}(u) = \sigma_\psi^2 + 2(1 - \rho_g) \sigma_\theta^2$, $V_l \equiv \text{Var}(l) = 2\rho_g \sigma_\theta^2$, and $V_u + V_l = \sigma_\psi^2 + 2\sigma_\theta^2$.

Step 1. The belief responds only to u_t^2 . Steady-state Bayesian consistency $\mathbb{E}[\hat{\sigma}_\psi^2] = \sigma_\psi^2$ pins the belief-to- u^2 proportionality at σ_ψ^2/V_u . The slope identified by (4) is therefore

$$\lambda_g = \frac{\sigma_\psi^2}{V_u} \cdot \frac{\text{Cov}(u_t^2, (u_t + l_t)^2)}{\text{Var}((u_t + l_t)^2)}.$$

Step 2. Under joint Gaussianity with $u \perp l$:

$$\text{Cov}(u^2, (u + l)^2) = \text{Var}(u^2) = 2V_u^2, \quad \text{Var}((u + l)^2) = 2(V_u + V_l)^2.$$

Step 3. Substitute:

$$\lambda_g = \frac{\sigma_\psi^2}{V_u} \cdot \frac{2V_u^2}{2(V_u + V_l)^2} = \frac{\sigma_\psi^2 V_u}{(V_u + V_l)^2} = \frac{\sigma_\psi^2 (\sigma_\psi^2 + 2(1 - \rho_g) \sigma_\theta^2)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2}.$$

This is (7).

Remark: the reading noise ν . The derivation abstracts from (5). Including it, the agent updates on $(u_t + \Delta \nu_t)^2$ while the regressor is unchanged; the slope becomes $\sigma_\psi^2 V_u^2 / [(V_u + 2\omega_g^2)(V_u + V_l)^2]$, i.e. (7) scaled by $V_u / (V_u + 2\omega_g^2) < 1$. The scaling is strongest where ω^2 is largest, so reading noise dampens the predicted λ at the bottom. The labelling channel must outweigh this in the data; the SMM of Section 9 disciplines the trade-off.

D Headline robustness

D.1: the headline with an occupation $\times$ time fixed effect added to the shock construction. D.2: one-at-a-time sweep over instrument lag depth, trim, and a COVID-excluded subsample. D.3: the sign-split shock.

D.1 Occupation $\times$ time fixed effect in the shock construction

To address the occupation concern of Section 7, I add an occupation $\times$ year-month fixed effect to the wage-shock construction. The interaction with time is the active term: a purely additive occupation effect cancels in the within-spell difference, since occupation is fixed within a stayer spell.

Group	Canonical			Occupation $\times$ time FE		
	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}_{\theta, \text{perc}, g}^2$	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}_{\theta, \text{perc}, g}^2$
Bot	0.628 (0.055)	+0.077 (0.030)	9.75 (1.71)	0.649 (0.063)	+0.091 (0.039)	7.20 (2.66)
Mid	0.912 (0.089)	+0.046 (0.015)	2.94 (0.93)	0.924 (0.091)	+0.036 (0.010)	2.74 (0.96)
Top	0.950 (0.060)	+0.026 (0.013)	4.12 (0.82)	0.957 (0.061)	+0.017 (0.008)	4.17 (0.74)
Wald μ (T–B)	$p = \mathbf{0.0001}$			$p = \mathbf{0.0004}$		
Wald λ (B–T)	$p = 0.118$			$p = 0.062$		
Wald σ^2 (B–T)	$p = \mathbf{0.003}$			$p = 0.274$		
Hansen J p (B/M/T)	0.20/0.23/0.24			0.64/0.14/0.44		

Table 7: Canonical headline (left) versus the same regression with an occupation $\times$ year-month fixed effect in the wage-shock construction (right). Standard errors in parentheses.

Read. Persistence and attribution gradients survive; the attribution gradient sharpens. The prior gradient holds in direction but loses cross-group significance: part of Bot’s inflated prior reads off cross-occupation transitory dispersion that the interaction removes. The labelling channel behind λ is robust to occupation controls; the reading-noise channel behind the prior operates partly through cross-occupation variation.

D.2 Lag depth, trim, and COVID exclusion

Each row of Table 8 varies one dial of the canonical specification: trim tightened to 2.5/97.5 within group, alternative instrument lag windows on each side, and a COVID-excluded subsample.

Variant	$\hat{\mu}_g$			$\hat{\lambda}_g$			$\hat{\sigma}_{\theta, \text{perc}, g}^2$			Wald p (B vs T)		
	B	M	T	B	M	T	B	M	T	μ	λ	σ^2
Canonical	0.628	0.912	0.950	+0.077	+0.046	+0.026	9.75	2.94	4.12	0.0001	0.118	0.003
Trim 2.5/97.5 within g	0.816	0.847	0.897	+0.059	+0.053	+0.014	11.23	4.07	4.86	0.110	0.352	0.000
$\bar{\sigma}^2$ lags 2–3	0.648	0.951	0.876	+0.075	+0.050	+0.020	9.96	2.85	4.44	0.000	0.079	0.003
$\bar{\sigma}^2$ lags 4–5	0.588	0.901	0.948	+0.102	+0.046	+0.021	9.03	3.00	4.38	0.000	0.061	0.032
Shock lags 0–1	0.683	0.937	0.940	+0.082	+0.016	+0.012	9.76	4.41	4.90	0.015	0.063	0.010
Shock lags 2–4	0.675	0.908	0.948	+0.095	+0.040	+0.027	8.76	3.20	4.10	0.003	0.008	0.011
Exclude 2020–21	0.636	0.995	0.886	+0.139	+0.043	+0.013	6.50	3.12	4.88	0.017	0.055	0.612

Table 8: One-at-a-time variations around the canonical headline. Bold p -values reject cross-group equality at the 5% level.

Read. The persistence and attribution gradients are robust across the lag variants, with the attribution Wald tighter than canonical in three of four. The within-group trim preserves the qualitative pattern but loosens both Walds: the tighter trim drops high-perceived-variance observations the canonical 1/99 retains. Excluding COVID sharpens the attribution gradient and breaks the prior gradient’s significance. The Hansen J never rejects.

D.3 Asymmetric extrapolation

Pooled across groups, the shock regressor is replaced by its sign-split components S^+ and S^- , with $S_{c,t}^+ = \mathbb{E}[(\Delta e_{i,t})^2 \cdot \mathbb{I}\{\Delta e_{i,t} > 0\} \mid c]$, S^- analogous, and $S^+ + S^- = S$ by construction. One λ^+ and one λ^- are identified jointly. Rows vary the cluster choice.

Cluster	$\hat{\mu}$	$\hat{\lambda}^+ (t)$	$\hat{\lambda}^- (t)$	$\hat{\lambda}^+ - \hat{\lambda}^-$	Wald p
2-way (respondent $\times$ quarter)	0.876 (0.050)	+0.055 (2.92)	+0.027 (1.59)	+0.028 (0.020)	0.163
state $\times$ group $\times$ quarter	0.876 (0.056)	+0.055 (3.48)	+0.027 (2.23)	+0.028 (0.018)	0.124
respondent only	0.876 (0.048)	+0.055 (4.89)	+0.027 (3.30)	+0.028 (0.016)	0.071

Table 9: Pooled asymmetric headline ($N = 20,363$, Hansen J $p = 0.11$). Bold marks 5% significance.

E Hours by income group

Both primitives of Section 8 run through hours. Fragmented, variable hours generate the reading noise ω_g^2 ; transitory shocks delivered through hours are the unlabelled kind, lowering ρ_g . Figure 5 documents the SIPP hours patterns.

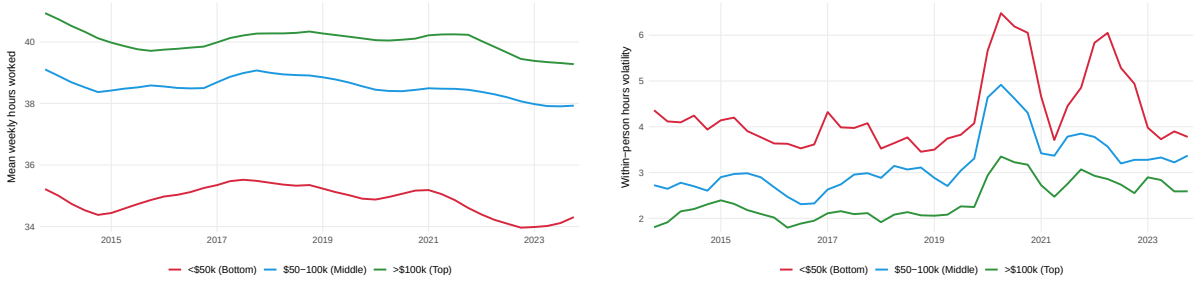


Figure 5: Mean weekly hours (left) and within-person hours volatility $\mathbb{E}[(\Delta \log h)^2 \mid g, t]$ in $(\%)^2$ (right) by income group. SIPP matched-stayer panel, quarterly with 4-quarter trailing MA. Hours volatility built from within-spell log differences, January seam dropped, 1%/99% trim.

F The update-speed channel and the form of the filter

The gain, attribution, and prior of Section 8 are the steady-state outputs of one filter, driven by the two pay-delivery primitives. This appendix gives the filter and the fit.

The signal. The agent's monthly signal is one squared pay reading, $z_t = (\Delta \tilde{y}_t)^2$ of (5), with mean $\sigma_\psi^2 + V_{x,g}$ and, under Gaussianity, variance $2(\sigma_\psi^2 + V_{x,g})^2$, where $V_{x,g} = 2(1 - \rho_g)\sigma_\theta^2 + 2\omega_g^2 = 2\sigma_{\theta, \text{perc}, g}^2$ is the contamination from unlabelled transitory pay and reading noise. The noise of a squared Gaussian scales with its level.

The gain. The agent smooths the attributed signal $\lambda_g z_t$ with weight μ_g . Treating the permanent variance as a random walk with drift q observed through this signal, the steady-state Kalman gain is

$$\mu_g = \frac{\bar{P}_g}{\bar{P}_g + R_g}, \quad R_g = \lambda_g^2 2(\sigma_\psi^2 + V_{x,g})^2, \quad \bar{P}_g = \frac{1}{2}(q + \sqrt{q^2 + 4qR_g}),$$

decreasing in R_g : the noisier the attributed signal she filters, the less each observation moves the belief. Substituting λ_g collapses the observation noise to $R_g = 2(\sigma_\psi^2)^2(V_u/S)^4$, with $V_u = \sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2$ the unlabelled residual variance, so the gain depends on the labelled share and not on reading noise, which cancels out of the signal-to-noise ratio. The bottom, labelling least, filters the noisiest unlabelled mix and updates slowest, the persistence gradient of Table 1.

The structural fit. The three belief moments per group follow from the two primitives and the common drift: the prior from (6), the attribution from (7), the gain from the filter above. Fitting (ω_g^2, ρ_g) and q to the estimated gains and priors in levels and the attribution in ratios (its level carries the cell-mean attenuation of Section 6) reproduces all three (Table 5), the top's gain included, at $J = 0.1$ on one degree of freedom. The fit drives the labelled share to its bounds, none of the bottom's transitory pay flagged and all of the top's; a milder labelling gradient leaves the bottom's attribution below the middle's, so the data pushes to the corners.

An additive variant. As a robustness check I also fit an additive-debias variant of the filter, which writes the prior as $(1 - \rho_g)\sigma_\theta^2 + \omega_g^2$, the gain as $\mu_g = \bar{P}_g/(\bar{P}_g + R_g)$, and the attribution slope from (7) scaled by the reading-noise factor $V_u/(V_u + 2\omega_g^2)$ of the remark to Appendix C (a noisier reader down-weights each signal, attenuating the slope), in cross-group ratios so the common cell-mean attenuation of Section 6 differences out. With ρ_g fixed to performance-pay incidence, (ω_g^2, q) are chosen to minimise $m(\theta)'\Omega^{-1}m(\theta)$, the standardised gap between these objects and their Table 4 estimates (Ω diagonal in the estimates' standard errors). It is not rejected ($J = 5.4$, 4 df, $p = 0.25$) but it undershoots μ_{Top} (a single drift cannot pair the top's high gain with its high prior, since a higher prior raises R_g and lowers the gain) and, because the large ω_{Bot}^2 over-attenuates Bot's slope, inverts the attribution gradient. Freeing ρ_g recovers the attribution direction only by driving ρ_g to the corners (0/0.9/1), and still misses the top's gain ($\chi^2(1) = 4.8$). The multiplicative rule of (2) avoids both, reproducing all three gradients including the top's from the filter (Table 5), so the household block of Section 9 reads its belief inputs from it.